Event Algebras and Applications to Cryptography

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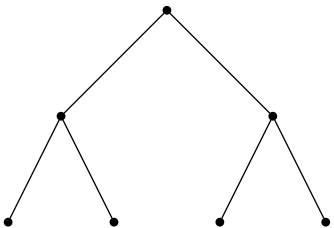
Outline

1. Events in discrete-step models
2. Event Algebras: operations and definition
3. Exactness of definition
4. Universal Event Inequalities (UEIs)
5. Cryptographically relevant statements as event inequalities:
   e.g. Hash-then-Sign security can be stated as

$$\mathsf{Forg} \preceq \mathsf{Forg}' \vee \mathsf{Coll}$$

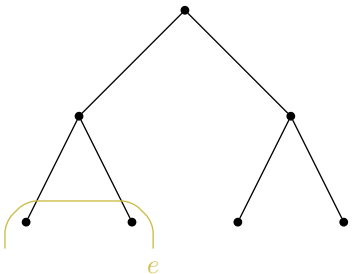
   "*An outer forgery implies either an inner forgery or a hash collision.*"
6. Summary

Discrete-step models define execution forests



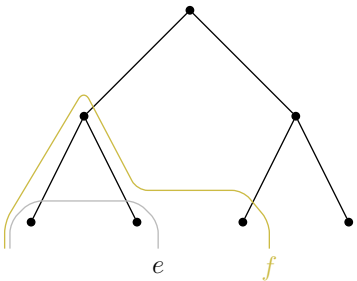
- Everywhere in CS: computation or actions proceed in discrete steps
- All possible sequences of steps form a tree (forest).
- An event is a set of nodes closed under taking steps: once it has occurred, it doesn't "unoccur".

Discrete-step models define execution forests



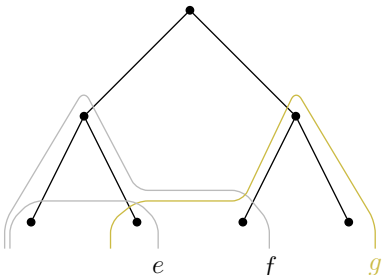
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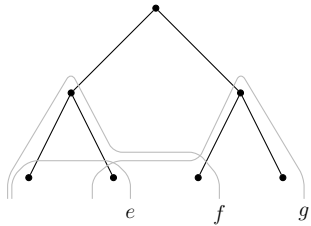
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Operations and order on events

Operations:

- $a \wedge b$: both have occurred.
- $a \vee b$: at least one has occurred.
- $a \doteq b$: a has occurred without b having occurred.
- $\top$: the sure event.
  $\bot$: the impossible event.

Order relation $a \preceq b$:
"a *cannot occur without* b".

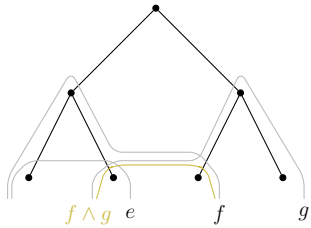


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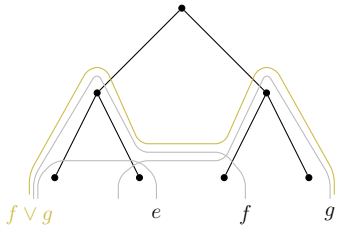


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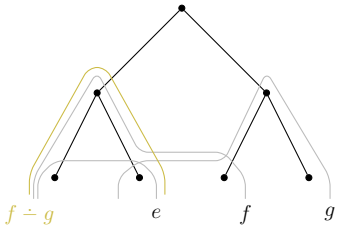


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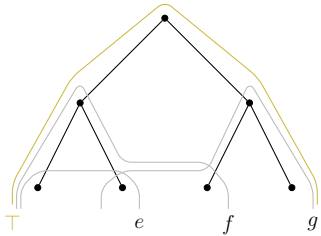


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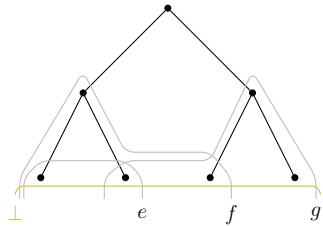


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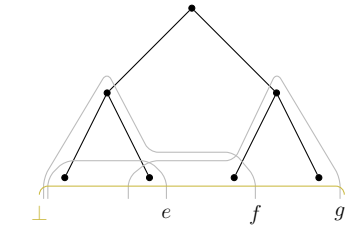


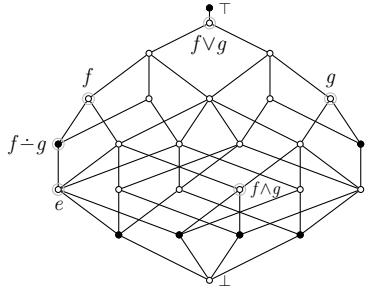
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All events of the above tree, ordered by $\preceq$.

Event algebras

Definition 1

An event algebra $(E; \preceq, \wedge, \vee, \dot{-}, \top, \bot)$ is a bounded lattice with a binary operation $\dot{-}$ such that for all $a, b, c \in E$:

$$
\begin{array}{ll}
\text{(E1)} & a \dot{-} a = \bot \\
\text{(E2)} & (a \dot{-} b) \vee a = a \\
\text{(E3)} & (a \dot{-} b) \vee b = a \vee b \\
\text{(E4)} & (a \vee b) \dot{-} c = (a \dot{-} c) \vee (b \dot{-} c) \\
\text{(E5)} & (a \dot{-} b) \wedge (b \dot{-} a) = \bot
\end{array}
$$

- (E5): histories are linearly ordered.
- Event algebras lie between bounded distributive lattices and Boolean algebras.

Exactness: event algebras are the algebras of events on forests

Forest poset P: every principal up-set $\uparrow x$ is a finite chain
(every node has a finite, linearly ordered past).

Theorem 1 (informal):

The events of every forest poset P form an event algebra.

Theorem 2 (via Birkhoff 1937):

For every finite event algebra E, the join-irreducibles $\mathsf{Ji}(E)$ form a forest poset, and $E \cong \mathsf{Dn}(\mathsf{Ji}(E))$.

$\Rightarrow$ (E1)–(E5) capture exactly the laws of events in discrete-step models.

Guarantees as (universal) event inequalities

- A bad event e usually cannot be ruled out ($e \neq \bot$).
  Instead, we prove

$$e \preceq f \qquad \text{“preventing } f \text{ prevents } e\text{”} \quad (\iff e \dot{-} f = \bot).$$

- A universal event inequality (UEI) holds in every event algebra:
  Example:

$$a \dot{-} b \preceq (a \dot{-} c) \vee (c \dot{-} b)$$

Advantages of guarantees as UEIs

- Model independence: by Thm. 1, a UEI holds in every forest.
  No computational model, adversary, or notion of efficiency.
- Composition:

$$e \preceq f \vee g \quad \text{and} \quad g \preceq h \vee i \quad \text{implies} \quad e \preceq f \vee h \vee i$$

- Probability: every σ-algebra is an event algebra. Hence

$$e \preceq f \vee g \quad \text{implies} \quad \Pr[e] \leq \Pr[f] + \Pr[g]$$

  for every probability measure.
- Purely algebraic proofs: well-suited for formalization in Lean/`mathlib`.

Example: tagging and hashing

Assume the following events are defined for all messages m, tags t, and hash values h:

- $\mathsf{Tag}_{\mathsf{A}}(m, t)$: party A has produced tag t for message m.
- $\mathsf{Acc}_{\mathsf{B}}(m, t)$: party B has accepted t as tag for m.
- $\mathsf{Hash}_{\mathsf{A}}(m, h)$, $\mathsf{Hash}_{\mathsf{B}}(m, h)$: A (resp. B) has computed h as the hash of m.

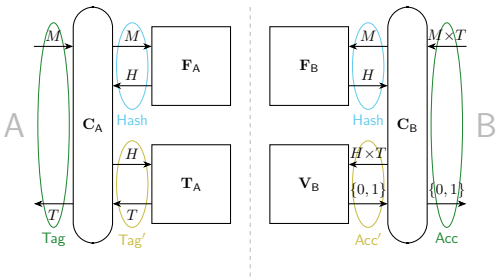
Forgery: B accepts a tag for m without A having tagged m:

$$\mathsf{Forg}_{\mathsf{A},\mathsf{B}} := \bigvee_{m,t} \left(\mathsf{Acc}_{\mathsf{B}}(m, t) \mathbin{\dot{-}} \bigvee_{t'} \mathsf{Tag}_{\mathsf{A}}(m, t') \right)$$

Collision: A and B have computed the same hash value for distinct messages:

$$\mathsf{Coll}_{\mathsf{A},\mathsf{B}} := \bigvee_{m \neq m',\, h} \left(\mathsf{Hash}_{\mathsf{A}}(m, h) \wedge \mathsf{Hash}_{\mathsf{B}}(m', h) \right)$$

Hash-then-Tag



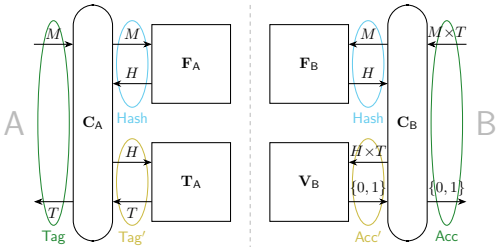
UEI for Hash-then-Tag:

$$\mathsf{Forg}_{\mathsf{A},\mathsf{B}} \preceq \mathsf{Coll}_{\mathsf{A},\mathsf{B}} \vee \mathsf{Forg}'_{\mathsf{A},\mathsf{B}} \vee \mathsf{S}_\mathsf{A} \vee \mathsf{S}_\mathsf{B}$$

"A forgery for the constructed scheme cannot occur without a hash collision, an inner forgery, or a protocol deviation of A or B."

No keys and no games: one UEI for Hash-then-Sign *and* Hash-then-MAC.

Hash-then-Tag



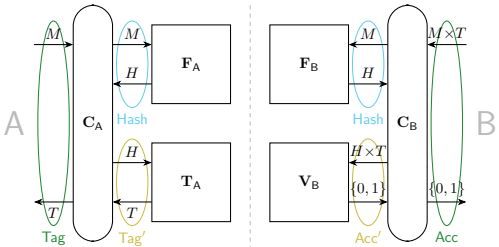
UEI for Hash-then-Tag:

$$\mathsf{Forg}_{\mathsf{A,B}} \preceq \mathsf{Coll}_{\mathsf{A,B}} \vee \mathsf{Forg}'_{\mathsf{A,B}} \vee \mathsf{S}_\mathsf{A} \vee \mathsf{S}_\mathsf{B}$$

"A forgery for the constructed scheme cannot occur without a hash collision, an inner forgery, or a protocol deviation of A or B."

$$\mathsf{S}_\mathsf{A} := \bigvee_{m,h,t} \left((\mathsf{Hash}_\mathsf{A}(m,h) \wedge \mathsf{Tag}'_\mathsf{A}(h,t)) \dot{-} \mathsf{Tag}_\mathsf{A}(m,t) \right)$$
$$\vee \bigvee_{h,t} \left(\mathsf{Tag}'_\mathsf{A}(h,t) \dot{-} \bigvee_{m} \mathsf{Hash}_\mathsf{A}(m,h) \right)$$

Hash-then-Tag



UEI for Hash-then-Tag:

$$\mathsf{Forg}_{\mathsf{A,B}} \preceq \mathsf{Coll}_{\mathsf{A,B}} \vee \mathsf{Forg}'_{\mathsf{A,B}} \vee \mathsf{S}_\mathsf{A} \vee \mathsf{S}_\mathsf{B}$$

"A forgery for the constructed scheme cannot occur without a hash collision, an inner forgery, or a protocol deviation of A or B."

$$\mathsf{S}_\mathsf{B} := \bigvee_{m,t} \left(\mathsf{Acc}_\mathsf{B}(m,t) \dot{-} \bigvee_{h} \left(\mathsf{Hash}_\mathsf{B}(m,h) \wedge \mathsf{Acc}'_\mathsf{B}(h,t) \right) \right)$$

HtT: formal statement

Lemma: For every event algebra E, finite sets X, Y, Z, and families of events

$$\mathsf{F}, \mathsf{G} : X \times Z \to E, \quad \mathsf{H}, \mathsf{I} : X \times Y \to E, \quad \mathsf{J}, \mathsf{K} : Y \times Z \to E,$$

we have

$$\begin{aligned}\bigvee_{x,z} \left(\mathsf{F}_{x,z} \dot{-} \bigvee_{z'} \mathsf{G}_{x,z'}\right) \preceq & \bigvee_{x \neq x', y} (\mathsf{H}_{x,y} \wedge \mathsf{I}_{x',y}) \vee \bigvee_{y,z} \left(\mathsf{J}_{y,z} \dot{-} \bigvee_{z'} \mathsf{K}_{y,z'}\right) \\ & \vee \bigvee_{x,y,z} \left((\mathsf{H}_{x,y} \wedge \mathsf{K}_{y,z}) \dot{-} \mathsf{G}_{x,z}\right) \vee \bigvee_{y,z} \left(\mathsf{K}_{y,z} \dot{-} \bigvee_{x} \mathsf{H}_{x,y}\right) \\ & \vee \bigvee_{x,z} \left(\mathsf{F}_{x,z} \dot{-} \bigvee_{y} (\mathsf{I}_{x,y} \wedge \mathsf{J}_{y,z})\right)\end{aligned}$$

More in the paper

Applications

- Transitivity of authentication, authentication amplification
- Authenticity from unforgeability
- Variants of unforgeability and their relations
- Strong unforgeability from one-time signatures [Bellare–Shoup, PKC'07]
- A signature combiner [Janneck, EC'26]

Full Version

- Infinite event algebras
- A formal event logic
- Free event algebras
- Correctness guarantees
- Feasibility of Lean/`mathlib` proofs

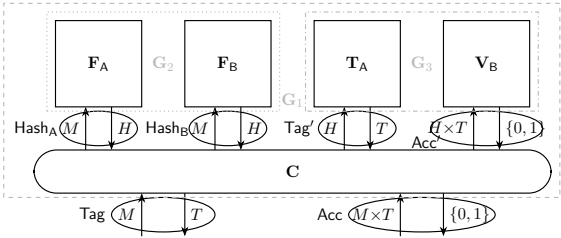
Summary

1. The executions of a discrete-step model form a forest, and define events. The axioms (E1)–(E5) capture exactly the laws of such events.
2. Many security guarantees are universal event inequalities: model-independent, composable, with purely algebraic proofs.
3. Hash-then-Tag is one UEI, for both signatures and MACs.

Outlook: UEIs for delivery guarantees, distributed protocols
Statements beyond UEIs: assumptions of the form "$a = \perp$ or $b = \perp$" (commitments), indistinguishability.

Thank you!

Backup: Relation to game-based proofs



Game-based : reductions $\mathcal{B}$ and $\mathcal{C}$ simulate $\mathbf{G}_1$ for an adversary $\mathcal{A}$.

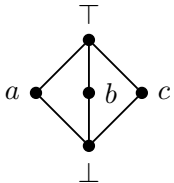
$\mathbf{G}_1$: forgery game of the constructed scheme (dashed)
$\mathbf{G}_2$: collision game (dotted), played by $\mathcal{B}$
$\mathbf{G}_3$: forgery game of the inner scheme (dash-dot), played by $\mathcal{C}$

Event-algebraic approach:

1. The executions of $\mathcal{A}$ with $\mathbf{G}_1$ form a forest, so the UEI holds. $\mathbf{G}_1$ follows the construction (S_A, S_B do not occur): $\mathsf{Win} \preceq \mathsf{Coll} \vee \mathsf{Forg}'$.
   This is the case distinction "$\mathsf{h}(m) = \mathsf{h}(m_i)$ for some queried m_i, or not".
2. Union bound: $\Pr[\mathsf{Win}] \leq \Pr[\mathsf{Coll}] + \Pr[\mathsf{Forg}']$, for every probability measure.

Reductions beyond UEIs: rewinding, imperfect simulation, indistinguishability.

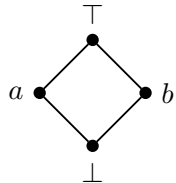
Backup: the axioms are independent



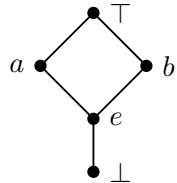
$\mathbf{M}_3$



$\mathbf{C}_2$



$\mathbf{B}_4$



$\mathrm{Dn}(P)$

- $\mathbf{M}_3$, $x \dot{-} y := \bot$ if $x \preceq y$, x if $y = \bot$, $\sigma(y)$ otherwise (σ cyclic on a, b, c): only (E2) fails.
- $\mathbf{C}_2$, $x \dot{-} y := \bot$: only (E3) fails.
- $\mathbf{B}_4$, $x \dot{-} y := \bot$ if $x \preceq y$, x otherwise: only (E4) fails.
- $\mathrm{Dn}(P)$ for $P = \{x \prec y, x \prec z\}$: only (E5) fails.
- (E1) follows from (E5) with $a = b$. It is kept so that (E1)–(E4) are the dual Heyting axioms.

Backup: why forests satisfy (E5)

Let P be a forest poset and $A, B \in \mathsf{Dn}(P)$. Suppose

$$x \in (A \dot{-}_P B) \cap (B \dot{-}_P A) = (A \setminus B)^{\downarrow} \cap (B \setminus A)^{\downarrow}.$$

- Then there are $y \in A \setminus B$ and $z \in B \setminus A$ with $x \leq y$ and $x \leq z$.
- $\uparrow x$ is a chain, so $y \leq z$ or $z \leq y$.
- $y \leq z$ and $z \in B$ give $y \in B$ (B is a down-set), a contradiction. Symmetrically for $z \leq y$.

(E1)–(E4) hold for the down-sets of *every* poset. (E5) is exactly where the forest structure enters.